

WN-GPT Agent

Our Multi-Agent Backend — Problem Statement, Flow, and Comparisons

Code: D:\Internship\WN_GPT\Backend

Not the same as wellnessgpt_tse.tex (HeyDoc WellnessGPT product)

This document describes *our agent* (internship implementation).

External comparators: Hippocratic AI, August AI

Related (separate) research files:

`wellnessgpt_tse.tex` = HeyDoc commercial WellnessGPT

`our_agent.tex` = Wellbeing (WB-001) requirements vision

Our roadmap: page-indexed retrieval to reduce hallucination and provide auditable guidance.

Document: WN-GPT Agent architecture research
Codebase: Backend/ — FastAPI + LangGraph + SQLite
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Abstract

This document describes **our agent**—the WN_GPT healthcare backend at **Backend/**—not the commercial **WellnessGPT** platform documented in `wellnessgpt_tse.tex` (HeyDoc AI). Our implementation is a **15-agent, LangGraph** system with **Azure OpenAI**, **SQLite**, and a **supervisor router** (`detect_intent`), not HeyDoc’s advertised **Agentic Orchestration Engine** or **HealthMem** stack. We state the problem, document the real process flow, contrast our agent with **Hippocratic AI** and **August AI**, and specify **page-indexed grounding** as our planned anti-hallucination layer.

1. Three Different Systems — Do Not Conflate

Three LaTeX artifacts in **Research/** refer to *related but distinct* systems. Treating them as one product causes incorrect architecture claims.

Name	File	What it is	Key architecture (claimed or built)
WellnessGPT (HeyDoc)	<code>wellnessgpt_tse.tex</code>	Commercial / marketing TSE for HeyDoc AI (wellness-gpt.com, HeyDoc app)	Three pillars: 15+ agents, Agentic Orchestration Engine , HealthMem (vector + graph + KV), HIPAA/FHIR/ABDM enterprise SaaS
Wellbeing (WB-001)	<code>our_agent.tex</code>	Software requirements / vision spec for a clinical Agentic OS	Hybrid RAG-first (Neo4j + Pinecone), LangGraph orchestrator, evidence panels, Epic/Cerner FHIR — target design , not identical to code
WN-GPT Agent (ours)	<code>Backend/ + this document</code>	Internship implementation you run locally	LangGraph supervisor router (no orchestrator LLM), 15 ReAct agents, SQLite only, local uploads; no HealthMem, no Neo4j/Pinecone in repo

Table 1: WellnessGPT (HeyDoc TSE) $\neq$ Wellbeing SRS $\neq$ WN-GPT Backend (our agent)

Rule for readers: When this repository says “WellnessGPT” in `main.py` or CLI branding, it is a *working title* for the demo backend. For research and comparisons, use **WN-GPT Agent** for our code and reserve **WellnessGPT** for `wellnessgpt_tse.tex` unless explicitly discussing HeyDoc’s product roadmap.

1.1. What Our Agent Actually Implements vs. HeyDoc Wellness-GPT TSE

Capability	wellnessgpt_tse.tex	WN-GPT
Orchestration	Full Agentic Orchestration Engine; multi-step agent handoffs	<code>detect_intent</code> + <code>plan</code>
Memory	HealthMem (vector + graph + KV, sub-150ms recall)	SQLite table
Compliance story	HIPAA, FHIR, ABDM production framing	Demo X-FHIR
RAG / evidence	Enterprise knowledge + EHR connectors	Tool calls to EHR
CLI / API	Enterprise deployment narrative	FastAPI + LLM

Table 2: Our agent is an early *prototype* of specialist routing, not the full HeyDoc WellnessGPT platform

2. Problem Statement (WN-GPT Agent)

2.1. Clinical and Operational Context

Healthcare organizations need AI that can support patients and staff across *many* workflows—triage, scheduling, pharmacy, insurance, nutrition, mental health, and hospital operations—without treating every question as a generic chat. Single-model chatbots fail because they lack persistent patient context, domain-specific tools, and traceable evidence for clinical statements.

2.2. Core Problems Addressed by Our Agent

1. **Fragmented patient journeys.** Users repeat symptoms, medications, and history across channels; there is no unified agent layer that routes requests to the correct clinical or administrative specialist.
2. **Hallucination in medical answers.** Large language models can produce confident but incorrect dosing, contraindication, or triage advice when not grounded in patient records or source documents.
3. **Lack of structured workflow execution.** Healthcare tasks (book appointment, log triage, file claim) require database writes and JSON-structured outputs, not free-form prose alone.
4. **India-specific interoperability.** Systems must align with ABHA/ABDM health records and local hospital operations, not only US-centric EHR copilots.
5. **Auditability and safety.** Clinicians and patients need to know *which agent* answered, *what data* was used, and *which document page* supports a recommendation.

2.3. Research Question

How can a multi-agent healthcare platform combine intent-based routing, tool-grounded specialists, and page-indexed retrieval to deliver safer, more explainable guidance than monolithic consumer health chatbots?

2.4. Scope of This Document

- **In scope:** Backend at Backend/ (API, LangGraph, agents, SQLite, CLI).
- **Out of scope:** Frontend UI, production HIPAA certification, live ABDM production APIs (stubs exist).
- **Planned (design):** Full page-index pipeline for uploaded PDFs and clinical corpora.

3. WN-GPT Backend Architecture (Our Implementation)

3.1. Repository Layout

Path	Role
main.py	FastAPI entry; CORS; auth middleware; mounts /api/v1 routers
app/core/graph_factory.py	Compiles LangGraph: intent node + 15 agents + utilities
app/services/routing.py	detect_intent (keywords + LLM) and route_to_agent
app/services/agents/	One Python module per specialist (ReAct + tools)
app/db/repository.py	SQLite CRUD (patients, triage, care plans, etc.)
app/schemas/state.py	Shared AgentState TypedDict for all nodes
cli/	Rich terminal UI; streaming demo (no LangGraph tools)
db_init/	Schema + seed SQL → wellness.db

Table 3: WN-GPT Agent backend structure (D:\Internship\WN_GPT\Backend)

3.2. Technology Stack

- **API:** FastAPI, Uvicorn, X-Patient-ID demo authentication
- **Orchestration:** LangGraph StateGraph, single-hop routing (no multi-agent chains)
- **LLM:** Azure OpenAI (e.g. GPT-4o) via LangChain
- **Agents:** create_react_agent with repository-backed @tool functions
- **Data:** SQLite (wellness.db); local uploads/ for documents

3.3. Fifteen Specialist Agents

Intent key	Graph node	Primary function
triage	triage_agent	Symptom urgency, specialty, emergency flag; logs to symptoms_logs
booking	booking_agent	Appointment scheduling, doctor slots
channeling	channeling_agent	Symptom → specialty routing
adherence	adherence_agent	Medication follow-up, reminders
care_plan	care_manager_agent	Preventive care, wellness plans
discharge	discharge_agent	Discharge summaries and instructions
insurance	insurance_agent	Coverage, claims, billing gaps
hospital_ops	hospital_ops_agent	Beds, inventory, ward status
pharmacy	pharmacy_agent	Prescriptions, interactions, refills

Intent key	Graph node	Primary function
mental_health	mental_health_agent	PHQ-9/GAD-7 screening, risk, escalation
family_care	family_care_agent	Household-level health aggregation
product_rec	product_rec_agent	Catalog-backed product suggestions
report_analysis	report_analyzer_agent	Lab/PDF report interpretation
nutrisense	nutrisense_agent	Diet plans, nutrition logs
fitguide	fitguide_agent	Exercise plans, fitness logs
health_records	health_records_node	ABDM/ABHA record pull (stub)
general	general_response	Fallback conversational LLM

Table 4: Agent registry in `graph_factory.py` (15 ReAct specialists + 2 utility nodes)

4. Agent Process Flow

4.1. Supervisor Routing (Not an Orchestrator Agent)

Our WN-GPT agent does **not** use HeyDoc’s “Agentic Orchestration Engine” nor a separate orchestrator LLM. It uses a **supervisor router** (`detect_intent`) that dispatches to exactly **one** specialist per user message—simpler than the multi-agent workflows described in `wellnessgpt_tse.tex`.

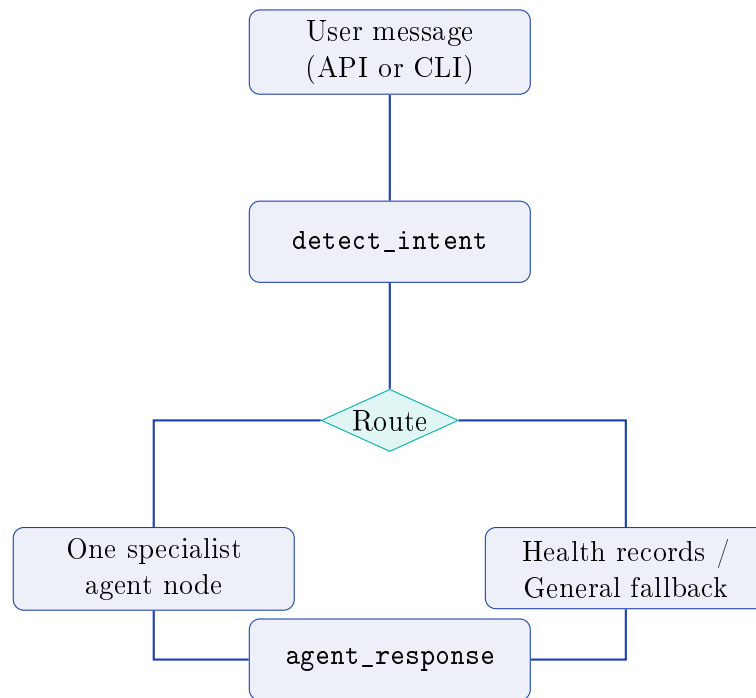


Figure 1: High-level LangGraph flow: single intent, single agent, then END

4.2. Intent Detection Algorithm

`detect_intent` in `routing.py` uses a **hybrid** strategy:

1. **Keyword pass** on the last user message (16 intent buckets: pain → triage, diet → nutrisense, etc.).
2. **LLM fallback** if no keyword matches: Azure OpenAI classifies into one intent name only.
3. **Normalization** via `normalize_enum` to reject invalid LLM labels.

4.3. Per-Agent Micro-Flow (API Path)

Each specialist agent follows the same ReAct pattern:

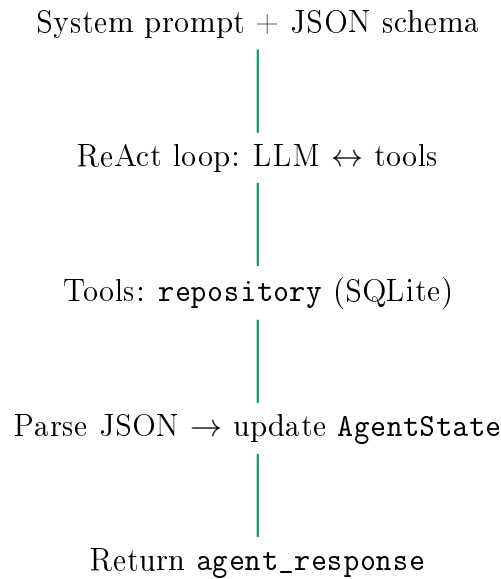


Figure 2: Inside one specialist agent (e.g. NutriSense, Triage, Pharmacy)

Example tools: `get_patient`, `get_triage_history`, `save_triage_result`, `get_nutrition_context`, `save_nutrition_plan`.

4.4. API vs CLI Paths

Aspect	API (POST /api/v1/chat/)	CLI (python -m cli)
Orchestration	Full LangGraph <code>graph.invoke</code>	<code>detect_intent</code> only
Agent tools	Yes (SQLite via ReAct)	No
Output	JSON <code>ChatResponse</code>	Rich streaming terminal
Patient auth	X-Patient-ID header	<code>-patient P001</code>

Table 5: Dual interfaces—research comparisons should use the API path for agent fidelity

4.5. Structured Data Side-Channels

Forms can bypass the graph and write directly to SQLite:

- POST /patient/{id}/symptoms, /nutrition, /fitness, /mental-health
- POST /patient/{id}/upload → local storage + `patient_documents`

5. Page-Indexed Grounding for Hallucination Reduction

5.1. Motivation

Keyword routing and SQLite tools reduce *wrong-agent* errors and *fabricated patient facts*, but they do not fully solve **document hallucination**—inventing lab values, misreading PDF tables, or citing non-existent guidelines. Consumer platforms (e.g. August AI) store reports in memory; enterprise voice agents (e.g. Hippocratic AI) use multi-model safety constellations; HeyDoc’s WellnessGPT TSE describes HealthMem + orchestration. **Our agent** proposes a complementary, implementable strategy: **page-indexed retrieval** on uploaded clinical PDFs (aligned with the RAG vision in `our_agent.tex`, but not yet built).

5.2. Page Indexing Concept

Page index: A structured map from each logical page (or section) of a clinical document to retrievable text, tables, figures, and metadata (document ID, page number, bounding region). Agents must cite (`doc_id`, `page`, `span`) when stating facts from uploads or corpora.

1. **Ingestion:** User uploads PDF/image via `/patient/{id}/upload` → stored under `uploads/`.
2. **Indexing:** Build a *pageindex* artifact: per-page text blocks, table HTML, figure paths (analogous to canonical bundle + pageindex pipelines used in document-AI systems).
3. **Retrieval:** On `report_analysis`, `pharmacy`, or `nutrisense` intents, fetch only top-*k* **pages** (not whole-document dumps) relevant to the query.
4. **Generation:** ReAct agent receives retrieved pages as context; system prompt requires answers to include **page references** and “insufficient evidence” when retrieval is empty.
5. **Guidance layer:** Map retrieved spans to patient-specific constraints (allergies, chronic conditions from SQLite) before final natural-language guidance.

5.3. Current vs Planned Implementation

Capability	Current backend	With page indexing
Patient profile grounding	Yes (SQLite tools)	Yes
Intent routing	Yes (keywords + LLM)	Yes
Document upload storage	Yes (uploads/)	Yes
Per-page index artifacts	No	Yes (planned)
Citation in agent output	No	Yes (planned)
Vector / semantic search	No	Optional hybrid with page filter
Report OCR	Stub/simulated	Real pipeline + page map

Table 6: Anti-hallucination layers: implemented today vs page-index roadmap

5.4. Why Page Indexing Reduces Hallucination

- **Bounded context:** LLM sees only retrieved pages, not entire 200-page PDFs (reduces confabulation from noise).
- **Verifiable citations:** UI can show “Page 3, Table 2” for clinician verification.
- **Deterministic anchors:** Tool calls return exact stored spans; model paraphrases *after* retrieval.
- **Agent-specific indexes:** NutriSense indexes dietary guidelines; Report Analyzer indexes lab PDFs; Pharmacy indexes prescription leaflets—each agent queries its allowed corpus.

6. Comparative Analysis

6.1. Platform Summaries

Hippocratic AI (USA, founded 2023) deploys **patient-facing voice agents** for chronic care outreach, intake, and administrative tasks. Its **Polaris constellation** uses a primary conversational model plus many specialized **safety engines** (overdose checks, escalation, medication reconciliation). Validation emphasizes **Real-World Evaluation (RWE-LLM)** with licensed clinicians reviewing outputs at scale.

August AI (India/global, founded 2022) is a **consumer health companion** (mobile app, WhatsApp) offering symptom guidance, secure storage of lab reports, and personalized tracking. It reports large user scale and Azure OpenAI infrastructure; it is **not** positioned as a 15-agent hospital OS but as a 24/7 personal health assistant.

WN-GPT Agent (our Backend/) is a **multi-agent LangGraph prototype** with 15 domain specialists, SQLite patient data, India-relevant ABDM stubs, and a planned **page-indexed** grounding layer. It is **not** the HeyDoc WellnessGPT enterprise platform from `wellnessgpt_tse.tex`.

6.2. Comparison Table

Dimension	Hippocratic AI	August AI	WN-GPT Agent (ours)
Primary user	Health systems, care teams	Individual consumers	Internship/demo; patients via API
Interaction mode	Voice-first out-bound/inbound	Chat, app, WhatsApp	REST API + Rich CLI
Agent architecture	Polaris multi-model constellation	Monolithic / unified companion	15 LangGraph ReAct specialists
Routing	Internal safety + task models	Intent within single product	<code>detect_intent</code> supervisor router
Memory	Call + EHR context	Long-term user memory, reports	SQLite only (not Health-Mem)
Grounding strategy	Many safety classifier models	Stored labs + conversation history	Tool SQL + page-indexed docs (planned)
Hallucination control	RWE-LLM, clinician review	Disclaimers; symptom checker	JSON schemas, tools, page cites (planned)
Clinical scope	Non-diagnostic outreach, admin	General wellness guidance	Triage, ops, insurance, nutrition, etc.
India / ABDM	US-focused	Strong India presence	ABHA/ABDM stub
Open backend	Proprietary enterprise	Proprietary consumer	<code>WN_GPT/Backend</code> (this repo)

Table 7: Hippocratic AI vs August AI vs **our** WN-GPT Agent (not HeyDoc WellnessGPT TSE)

6.3. Process Flow Comparison

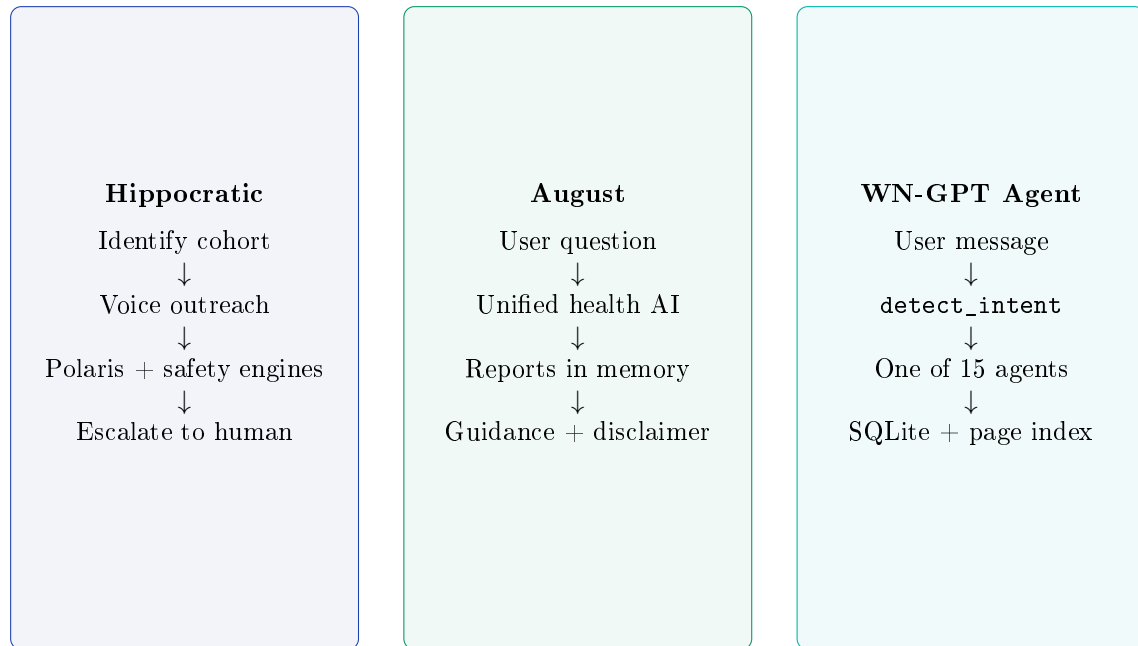


Figure 3: Simplified operational flows (conceptual)

6.4. Strategic Positioning (Our Agent)

- **vs Hippocratic AI:** Our agent prototypes broader in-app workflows (beds, insurance, nutrisense), not voice outreach; page indexing targets document verifiability with explicit page citations.
- **vs August AI:** Our agent trades consumer simplicity for specialist routing and hospital ops APIs; page-indexed retrieval is the planned answer to lab-report grounding.
- **vs HeyDoc WellnessGPT (wellnessgpt_tse.tex):** HeyDoc describes enterprise orchestration + HealthMem; we implement a **minimal viable** multi-agent router in SQLite as a step toward that vision.
- **vs Wellbeing SRS (our_agent.tex):** WB-001 specifies Neo4j/Pinecone RAG; our code is the **first engineering slice**—LangGraph agents + planned page index instead of full hybrid memory.
- **Our thesis:** Specialist agents + supervisor routing + page-indexed evidence → safer guidance without claiming HealthMem or full orchestration yet.

7. Conclusion and Future Work

7.1. Summary

The **WN-GPT Agent** backend implements a **15-agent, single-hop LangGraph** architecture with hybrid intent detection and SQLite-grounded ReAct tools. It is **not** the HeyDoc WellnessGPT platform in `wellnessgpt_tse.tex` (no HealthMem, no enterprise orchestration engine). Recommended evolution: **page-indexed retrieval** on `report_analyzer_agent`, pharmacy, and nutrisense, with mandatory page citations—bridging toward the Wellbeing RAG vision in `our_agent.tex`.

7.2. Future Work

1. Implement pageindex builder for `patient_documents` uploads.
2. Add retrieval tools: `search_pages(query, doc_id, k)` to agent toolkits.
3. Extend CLI to invoke LangGraph (parity with API).
4. Profile update API (allergies, conditions) with audit log.
5. Replace ABDM/OCR stubs with production connectors.
6. Evaluate hallucination rate with/without page index on held-out lab PDFs.

7.3. References

- Hippocratic AI — Polaris, Safety, RWE-LLM: <https://hippocraticai.com>
- August AI — <https://www.meetaugust.ai>
- HeyDoc WellnessGPT TSE (separate product doc): `wellnessgpt_tse.tex`
- Wellbeing WB-001 SRS (requirements vision): `our_agent.tex`
- **Our implementation:** `D:\Internship\WN_GPT\Backend`